

AI Model with Track Map and SLAM Navigation

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- 1. Course Content
- 2. Preparation Work
- 3. Run Case Study
 - 3.1 Start Program
 - 3.2 Test Case

1. Course Content

1. Combine prerequisite AI large model functionality to use robot's AI visual understanding and SLAM navigation for comprehensive case studies in the track map. This course requires the use of the accompanying track map

Note: For a better experience, please first learn and master the following content in the product tutorial:

- AI Large Model: [04. AI Model Development], [05.AI Model-Voice Version]—[3.Multimodal visual understanding+SLAM navigation]
- Road Network Planning Navigation: [07.Road Network Planning and Navigation]

2. Preparation Work

- Use the following command to start the microROS agent and connect to the chassis

[!WARNING]

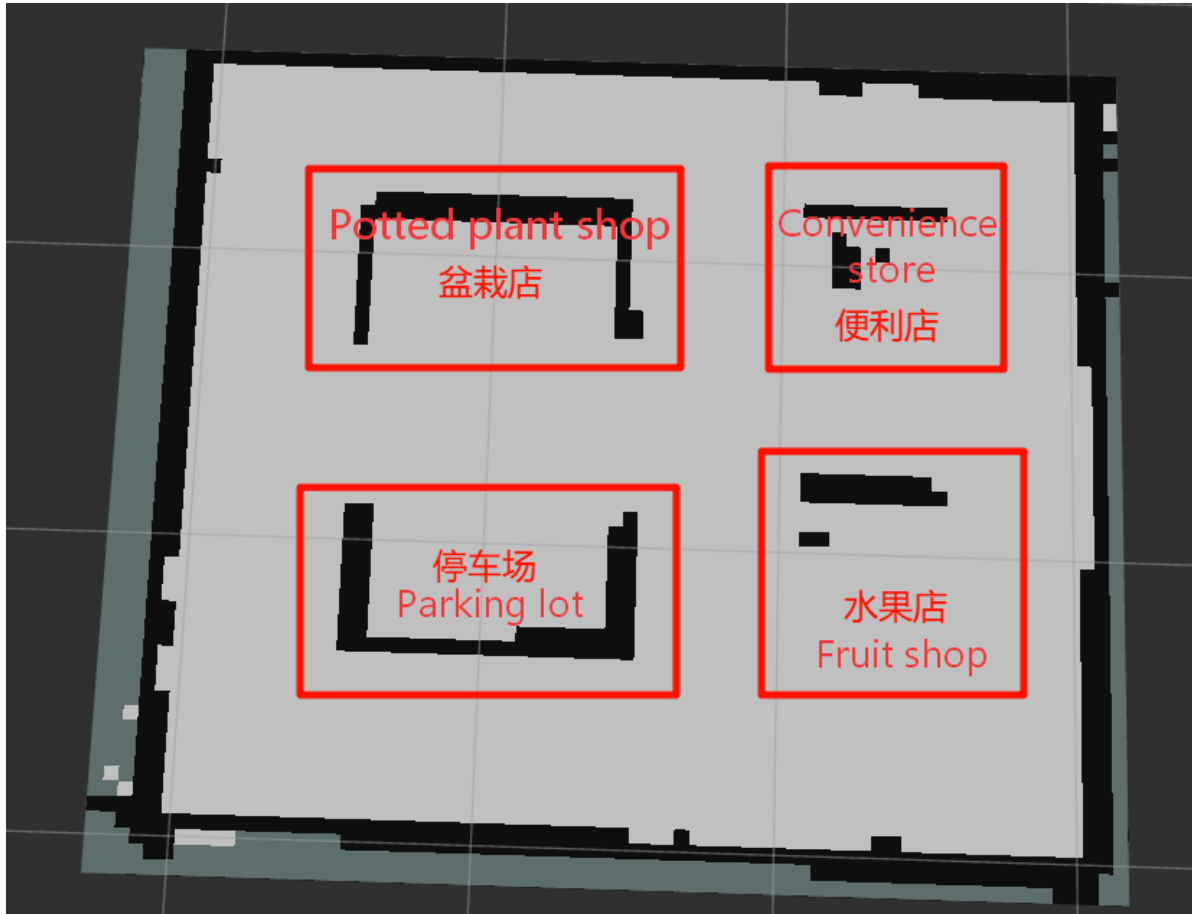
If the agent is already started, there is no need to start it again

```
start_agent
```

```
jetson@yahboom: ~
jetson@yahboom: ~ 80x24
ubsubscriber created | client_key: 0x14F22590, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1763535321.445792] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x14F22590, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1763535321.449556] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x14F22590, topic_id: 0x006(2), participant_
id: 0x000(1)
[1763535321.452876] info | ProxyClient.cpp | create_subscriber | s
ubsubscriber created | client_key: 0x14F22590, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1763535321.457712] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x14F22590, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1763535321.461959] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x14F22590, topic_id: 0x007(2), participant_
id: 0x000(1)
[1763535321.466141] info | ProxyClient.cpp | create_subscriber | s
ubsubscriber created | client_key: 0x14F22590, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1763535321.473112] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x14F22590, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```



We can name any point on the map. Here is an example of naming as shown in the figure



- **For methods to obtain vehicle target points on the map**, please refer to [05.AI Model-Voice Version]—[3.Multimodal visual understanding+SLAM navigation—2.3 Configure Map Mapping File]. The method is completely the same. This course is a reference scenario with the track map
- Edit the map mapping file (any editing method can be used)

```
~/yahboomM3_ws/user_ws/src/m3_bringup/config/multi_brains/map_mapping.yaml
```

- Regular navigation points need to be filled into the points under the `common_map_areas` field. Add or delete according to your needs

```
common_map_areas: #Common navigation
A:
  name: 'School'
  position:
    x: 2.315
    y: 1.444
    z: 0.0
  orientation:
    x: 0.0
    y: 0.0
    z: 0.673
    w: 0.740
B:
  name: 'Library'
```

```
position:
  x: 2.797
  y: 0.165
  z: 0.0
orientation:
  x: 0.0
  y: 0.0
  z: -0.627
  w: 0.779
```

- Additionally, you need to update the new map mapping information in Dify and publish save settings. For configuration methods, refer to [05.AI Model-Voice Version]—[3.Multimodal visual understanding+SLAM navigation—2.4 Configure Map Mapping Variables in Dify]

Edit Conversation Variable

Name
map_mapping

Type
array[string]

Default Value [Edit in JSON](#)

School: A

Library: B

+ Add Value

Description
Map mapping stores locations that need to be known in advance by AI

Cancel Save

[!TIP]

⚠ For the same test command, the large model's response content will not be completely identical and will differ slightly from the screenshots in the tutorial. If you need to enhance or reduce the diversity of large model responses, refer to the [4.AI Model Development] - [05.AI Model Development—Chatbot] course section on configuring decision layer large

model parameters.

3. Run Case Study

3.1 Start Program

- Start AI large model functionality on the vehicle terminal, or you can use the shortcut command `multi_brains`

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

Open a container terminal,

```
exec_m3
```

```
ros2 launch m3_bringup llm_agent_control.launch.py
```

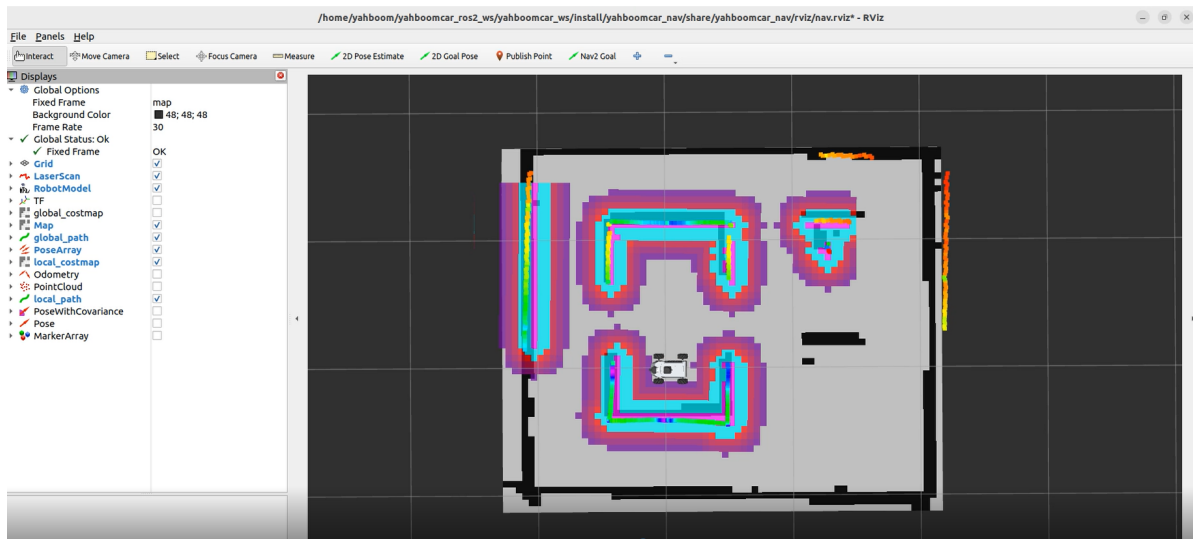
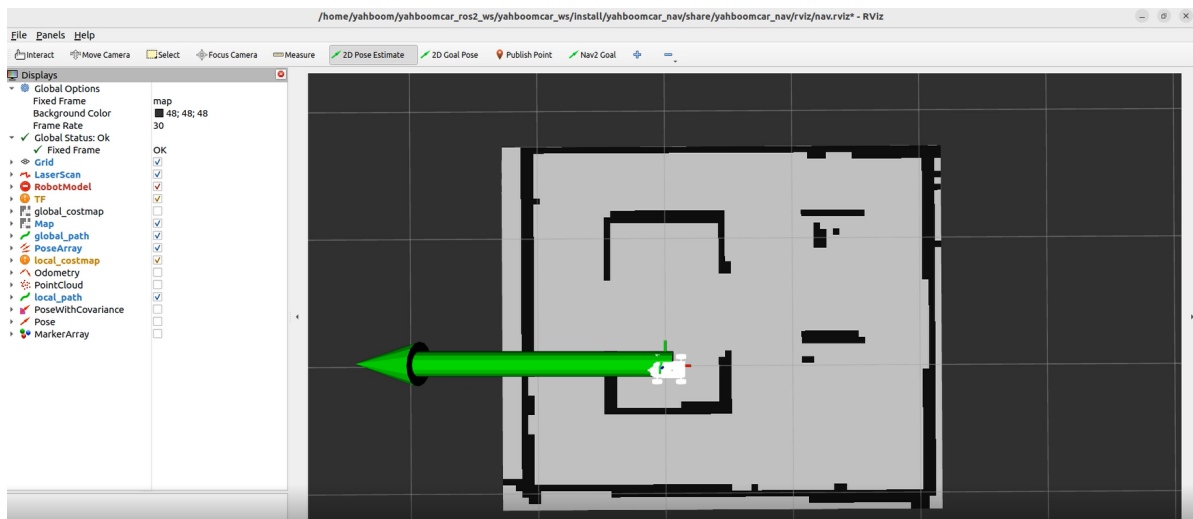
After initialization is complete, the following content will be displayed

```
[action_service_nuwa-15] from pkg_resources import resource_stream, resource_exists
[action_service_nuwa-15] [INFO] [1766129486.356630338] [action_service_nuwa]: enable_route_nav: False
[action_service_nuwa-15] [INFO] [1766129486.726464447] [action_service_nuwa]: ROS Action Service Initialization completed
[model_service-14] [INFO] [1766129518.715920692] [model_service]: Dify LLM connection successful
[model_service-14] [WARN] [1766129518.719145287] [rcl.logging_rosout]: Publisher already registered for provided node name. If this is due to multiple nodes with the same name then all logs for that logger name will go out over the existing publisher. As soon as any node with that name is destructed it will unregister the publisher, preventing any further logs for that name from being published on the rosout topic.
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.rear
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.center_lfe
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.side
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.hdmi
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.hdmi
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[model_service-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[model_service-14] ALSA lib pcm_oss.c:397:(snd_pcm_oss_open) Cannot open device /dev/dsp
[model_service-14] ALSA lib pcm_oss.c:397:(snd_pcm_oss_open) Cannot open device /dev/dsp
[model_service-14] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card
[model_service-14] ALSA lib pcm_usb_stream.c:482:(snd_pcm_usb_stream_open) Invalid card 'card'
[model_service-14] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card
[model_service-14] ALSA lib pcm_usb_stream.c:482:(snd_pcm_usb_stream_open) Invalid card 'card'
[model_service-14] [INFO] [1766129519.225800065] [ASR_Detect]: The online ASR Engine:paraformer-realtime-v2 initialization has been completed
[model_service-14] [INFO] [1766129519.274575027] [model_service]: Tongyi TTS Engine initialized successfully
[model_service-14] [INFO] [1766129519.277227745] [model_service]: ROS LLM Service Initialization completed
```

- Start navigation node (uses AMCL localization algorithm by default)

```
ros2 launch m3_bringup nav2.launch.py
```

Then follow the navigation function startup process to initialize positioning. The rviz2 visualization interface will open. Click **2D Pose Estimate** in the top toolbar to enter selection state, roughly mark the robot's position and orientation on the map. After initializing positioning, the preparation work is complete



3.2 Test Case

Here are reference test cases. Users can create their own dialogue commands

Please first remember your current position, then go to the fruit store and convenience store in sequence, return to your original position, and tell me what you saw.

First use "Hello yahboom" to wake up the robot. After the robot responds and enters recording state, users can speak. The robot will perform dynamic sound detection. If there is sound activity, it will print 1-1-1-1, if no sound activity it will print ----- . When speaking ends, tail sound detection will be performed. If silence exceeds 1 second, recording will stop. Dynamic sound detection (VAD) is shown in the figure below:


```
sunrise@ubuntu: ~  
sunrise@ubuntu: ~ 125x41  
[action_service usb-13] from pkg resources import resource, resource_exists  
[imu_filter madgwick node-6] [INFO] [1760406802.016424537] [imu_filter madgwick]: First IMU message received.  
[joint_state publisher-2] [INFO] [1760406802.045628686] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description topic...  
[asr-14] [INFO] [1760406845.891056276] [asr]: The online asr model :gummy-chat-v1 is loaded  
[asr-14] [INFO] [1760406845.929901140] [asr]: asr_node Initialization completed  
[action_service usb-13] [INFO] [1760406846.994750362] [action_service]: action service started...  
[model_service-12] [INFO] [1760406848.181021930] [model_service]: LargeModelService node Initialization completed...  
[asr-14] [INFO] [1760406985.899403811] [asr]: I'm here  
[asr-14] ALSA lib pcm dmix.c:1032:(snd_pcm_dmix open) unable to open slave  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.rear  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.center_lfe  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.side  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.hdmi  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.hdmi  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phonein  
[asr-14] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phonein  
[asr-14] ALSA lib pcm_oss.c:397:(snd_pcm_oss_open) Cannot open device /dev/dsp  
[asr-14] ALSA lib pcm_oss.c:397:(snd_pcm_oss_open) Cannot open device /dev/dsp  
[asr-14] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card  
[asr-14] ALSA lib pcm_usb_stream.c:482:(snd_pcm_usb_stream open) Invalid card 'card'  
[asr-14] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card  
[asr-14] ALSA lib pcm_usb_stream.c:482:(snd_pcm_usb_stream open) Invalid card 'card'  
[asr-14] ALSA lib pcm dmix.c:1032:(snd_pcm_dmix open) unable to open slave  
[asr-14] [INFO] [1760406987.880841151] [asr]: 1  
[asr-14] [INFO] [1760406987.944597697] [asr]: 1  
[asr-14] [INFO] [1760406988.004579198] [asr]: 1  
[asr-14] [INFO] [1760406988.061712031] [asr]: 1  
[asr-14] [INFO] [1760406988.124015326] [asr]: 1  
[asr-14] [INFO] [1760406988.193104750] [asr]: -  
[asr-14] [INFO] [1760406988.253492544] [asr]: -  
[asr-14] [INFO] [1760406988.314953421] [asr]: -  
[asr-14] [INFO] [1760406988.376768924] [asr]: -  
[asr-14] [INFO] [1760406988.438891385] [asr]: -  
[asr-14] [INFO] [1760406988.475412495] [asr]: -  
[asr-14] [INFO] [1760406988.538105831] [asr]: -  
[asr-14] [INFO] [1760406988.599228750] [asr]: -  
[asr-14] [INFO] [1760406988.662174045] [asr]: -  
[asr-14] [INFO] [1760406988.721234796] [asr]: -
```

The robot will first reply to the user, then execute actions according to the command, while printing information in the terminal as follows:

```
Applications sunrise@ubuntu: ~/Desktop  
sunrise@ubuntu: ~/Desktop 125x41  
[asr-14] [INFO] [1760409924.594647191] [asr]: -  
[asr-14] [INFO] [1760409924.633898613] [asr]: -  
[asr-14] [INFO] [1760409924.702939117] [asr]: -  
[asr-14] [INFO] [1760409924.769259147] [asr]: -  
[asr-14] [INFO] [1760409924.828899211] [asr]: -  
[asr-14] [INFO] [1760409924.893134188] [asr]: -  
[asr-14] [INFO] [1760409924.954098207] [asr]: -  
[asr-14] [INFO] [1760409925.019916889] [asr]: -  
[asr-14] [INFO] [1760409925.063276862] [asr]: -  
[asr-14] [INFO] [1760409925.122269396] [asr]: -  
[asr-14] [INFO] [1760409925.181049650] [asr]: -  
[asr-14] [INFO] [1760409925.245269395] [asr]: -  
[asr-14] [INFO] [1760409925.308561452] [asr]: -  
[asr-14] [INFO] [1760409925.370602235] [asr]: -  
[asr-14] [INFO] [1760409925.435111034] [asr]: -  
[asr-14] [INFO] [1760409925.465799757] [asr]: -  
[asr-14] [INFO] [1760409925.528144423] [asr]: -  
[asr-14] [INFO] [1760409925.591287725] [asr]: -  
[asr-14] [INFO] [1760409925.651482296] [asr]: -  
[asr-14] [INFO] [1760409925.715464558] [asr]: -  
[asr-14] [INFO] [1760409925.776213409] [asr]: -  
[asr-14] [INFO] [1760409925.837235527] [asr]: -  
[asr-14] [INFO] [1760409928.747220973] [asr]: 请你先记住现在的位置，然后依次去水果店和便利店返回到原来的位置，告诉我你都看到了什么东西。  
[asr-14] [INFO] [1760409928.750486364] [asr]: @okay, let me think for a moment...  
[model_service-12] [INFO] [1760409935.823681474] [model_service]: 决策层AI规划:1. 调用记录当前位置函数，记录位置  
[model_service-12] 2. 导航到水果店  
[model_service-12] 3. 调用视觉函数观察周围环境，并记录观察结果  
[model_service-12] 4. 导航到便利店  
[model_service-12] 5. 调用视觉函数观察周围环境，并记录观察结果  
[model_service-12] 6. 返回初始位置  
[model_service-12] 7. 将在水果店和便利店观察到的物品告知用户  
[model_service-12] [INFO] [1760409942.339012741] [model_service]: "action": ['get_current_pose()'], "response": 好的，我现在记住当前位置了。接下来我会去水果店和便利店看看，并告诉你我看到了什么。  
[action_service usb-13] [INFO] [1760409951.745392583] [action_service]: Recorded Pose - Position: x=0.05996864206231826, y=-0.001885812804798556, z=0.0, Orientation: x=0.0, y=0.0, z=0.9999999663923336, w=-0.00025925919808082477  
[action_service usb-13] [INFO] [1760409951.757109709] [action_service]: Published message: 机器人反馈:get_current_pose()成功  
[model_service-12] [INFO] [1760409953.821517973] [model_service]: "action": ['navigation(A)'], "response": 我已经记住了当前位置，现在准备导航去水果店。
```

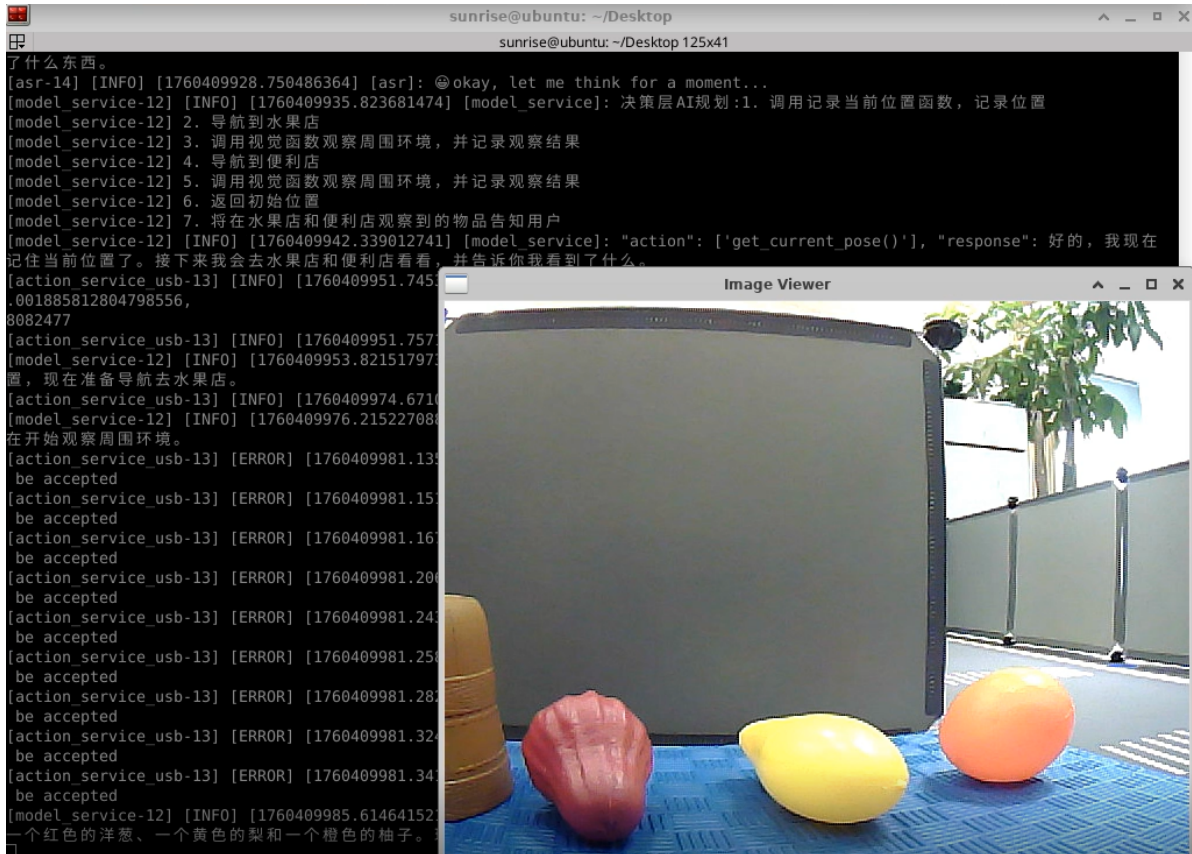
Decision layer large model outputs planned task steps:

```

[model_service-12] [INFO] [1760409935.823681474] [model_service]: 决策层AI规划:1. 调用记录当前位置函数, 记录位置
[model_service-12] 2. 导航到水果店
[model_service-12] 3. 调用视觉函数观察周围环境, 并记录观察结果
[model_service-12] 4. 导航到便利店
[model_service-12] 5. 调用视觉函数观察周围环境, 并记录观察结果
[model_service-12] 6. 返回初始位置
[model_service-12] 7. 将在水果店和便利店观察到的物品告知用户

```

After that, the execution layer large model will execute according to these task steps:



When the task is completed, use "Hello,yahboom" to wake up the robot, the robot responds: "I'm here, please give instructions", at this time you can give the next command


```
sunrise@ubuntu: ~/Desktop
sunrise@ubuntu: ~/Desktop 125x41
[action_service_usb-13] [ERROR] [1760409981.282069568] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760409981.324090812] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760409981.341005415] [action_service]: The image is being saved and no new information will be accepted
[model_service-12] [INFO] [1760409985.614641521] [model_service]: "action": ['navigation(C)'], "response": 在水果店，我看到了一个红色的洋葱、一个黄色的梨和一个橙色的柚子。现在我将前往便利店。
[action_service_usb-13] [INFO] [1760410009.050422579] [action_service]: Published message: 机器人反馈:执行navigation(C)完成
[model_service-12] [INFO] [1760410011.116937432] [model_service]: "action": ['seewhat()'], "response": 我已经到达便利店，现在开始观察周围环境。
[action_service_usb-13] [ERROR] [1760410015.760043930] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760410015.776716037] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760410015.817135095] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760410015.838184189] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760410015.860054990] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760410015.896011637] [action_service]: The image is being saved and no new information will be accepted
[action_service_usb-13] [ERROR] [1760410015.922096241] [action_service]: The image is being saved and no new information will be accepted
[model_service-12] [INFO] [1760410019.950384394] [model_service]: "action": ['navigation(zero)'], "response": 在便利店，我看到一个篮子里有蓝色和红色的物品。现在我将返回到原来的位置。
[action_service_usb-13] [INFO] [1760410053.650422781] [action_service]: Navigation failed with status: 6
[action_service_usb-13] [INFO] [1760410053.656257538] [action_service]: Published message: 机器人反馈:执行navigation(zero)失败
[model_service-12] [INFO] [1760410055.950689589] [model_service]: "action": ['navigation(zero)'], "response": 哎呀，返回原来位置时遇到了一点小问题，让我再试一次。
[action_service_usb-13] [INFO] [1760410093.548177663] [action_service]: Published message: 机器人反馈:执行navigation(zero)完成
[model_service-12] [INFO] [1760410097.636336494] [model_service]: "action": [], "response": 我已经成功返回到原来的位置。让我来告诉你我在水果店和便利店看到的東西：在水果店，我看到一个红色的洋葱、一个黄色的梨和一个橙色的柚子；在便利店，我看到一个篮子里有蓝色和红色的物品。这就是我的观察结果哦！
[action_service_usb-13] [INFO] [1760410119.507591328] [action_service]: Published message: 机器人反馈:回复用户完成
[model_service-12] [INFO] [1760410121.481314996] [model_service]: "action": ['finishtask()'], "response": 我已经完成了全部任务，如果你还有其他需要帮忙的，请随时告诉我！
```

After the robot completes the task, it will enter a waiting state. At this time, the robot enters free dialogue state again, but all dialogue history will be retained. You can wake up yahboom again, "end current task", to let the robot end the current task cycle, clear the dialogue history, and start a new task cycle.

```
jetson@yahboom: ~ 122x34
[asr-14] [INFO] [1755679669.327887218] [asr]: 1
[asr-14] [INFO] [1755679669.388356288] [asr]: 1
[asr-14] [INFO] [1755679669.448540247] [asr]: 1
[asr-14] [INFO] [1755679669.509674586] [asr]: 1
[asr-14] [INFO] [1755679669.571171770] [asr]: 1
[asr-14] [INFO] [1755679669.630793924] [asr]: -
[asr-14] [INFO] [1755679669.691181227] [asr]: -
[asr-14] [INFO] [1755679669.751568947] [asr]: -
[asr-14] [INFO] [1755679669.812568907] [asr]: -
[asr-14] [INFO] [1755679669.873032793] [asr]: -
[asr-14] [INFO] [1755679669.933673172] [asr]: -
[asr-14] [INFO] [1755679669.994094078] [asr]: -
[asr-14] [INFO] [1755679670.024206570] [asr]: -
[asr-14] [INFO] [1755679670.084441603] [asr]: -
[asr-14] [INFO] [1755679670.145278316] [asr]: -
[asr-14] [INFO] [1755679670.205037150] [asr]: -
[asr-14] [INFO] [1755679670.266176063] [asr]: -
[asr-14] [INFO] [1755679670.326090750] [asr]: -
[asr-14] [INFO] [1755679670.386630159] [asr]: -
[asr-14] [INFO] [1755679670.446504875] [asr]: -
[asr-14] [INFO] [1755679670.506955765] [asr]: -
[asr-14] [INFO] [1755679670.567423776] [asr]: -
[asr-14] [INFO] [1755679670.628749552] [asr]: -
[asr-14] [INFO] [1755679670.688834205] [asr]: -
[asr-14] [INFO] [1755679670.750200240] [asr]: -
[asr-14] [INFO] [1755679670.809695021] [asr]: -
[asr-14] [INFO] [1755679670.870388459] [asr]: -
[asr-14] [INFO] [1755679671.689391598] [asr]: 结束当前任务。
[asr-14] [INFO] [1755679671.691073141] [asr]: 😊okay, let me think for a moment...
[model_service-12] [INFO] [1755679673.748230368] [model_service]: "action": ['finish_dialogue()'], "response": 好的，任务已经结束了，我就退下休息啦，有需要再叫我哦~
[model_service-12] [INFO] [1755679679.725228752] [model_service]: The current instruction cycle has ended
[action_service_usb-13] [INFO] [1755679679.725376348] [action_service]: Published message: finish
```

